Complex Mapping Analysis

Geometric Inversion of a Circle under $w = 1/z$

Abstract

This report investigates the geometric properties of the complex inversion mapping, $w = 1/z$. Specifically, it analyzes the transformation of a circle passing through the origin in the complex z-plane. Both analytical derivation and numerical MATLAB simulations are presented to demonstrate that such a circle is mapped to a straight line in the w-plane.

1 Mathematical Formulation

The complex transformation $w = f(z) = \frac{1}{z}$ performs a geometric inversion in the complex plane. Let $z = x + iy$ and $w = u + iv$. The mapping relates the coordinates as follows:

$$w = \frac{1}{x+iy} = \frac{x-iy}{x^2+y^2} \implies u = \frac{x}{x^2+y^2}, \quad v = \frac{-y}{x^2+y^2} \tag{1}$$

Since the transformation is an involution, the inverse mapping is identical: $z = \frac{1}{w}$. Therefore:

$$x = \frac{u}{u^2+v^2}, \quad y = \frac{-v}{u^2+v^2} \tag{2}$$

1.1 Transformation of the Given Circle

Consider the circle defined in the z-plane by the equation:

$$(x+2)^2 + (y-2)^2 = 8 \tag{3}$$

Expanding this equation yields:

$$x^2 + y^2 + 4x - 4y = 0 \tag{4}$$

Notice that the constant term is zero, confirming that this circle passes through the origin $(0, 0)$. To find the image of this circle in the w-plane, we substitute x and y using the inverse mapping relations:

$$\frac{1}{u^2+v^2} + 4\left(\frac{u}{u^2+v^2}\right) - 4\left(\frac{-v}{u^2+v^2}\right) = 0 \tag{5}$$

Multiplying the entire equation by $u^2 + v^2$ (assuming $w \neq 0$) simplifies the expression to:

$$1 + 4u + 4v = 0 \implies v = -u - 0.25 \tag{6}$$

This represents a linear equation in the w-plane. Thus, we have analytically proven that a circle passing through the origin in the z-plane maps to a straight line in the w-plane under the inversion mapping.

2 Computational Implementation

To visualize this conformal mapping, a MATLAB script employs the `contour` function. Instead of plotting discrete points, the code evaluates the original equation over a discrete grid Z and evaluates the transformed geometry by mapping a W grid back to the Z-plane domain.

```
% Create a grid for the W-plane
[U, V] = meshgrid(linspace(-10, 10, 600));
W = U + 1i * V;

% Apply inverse mapping z = 1/w
Z_from_W = 1 ./ W;

% Evaluate the circle equation to find the image contour
W_image = (real(Z_from_W) + 2).^2 + (imag(Z_from_W) - 2).^2 - 8;

% Plot the zero-level set (the mapped line)
contour(U, V, W_image, [0 0]);
```

Listing 1: MATLAB Implementation of the Transformation

The resulting plots confirm the analytical derivation, showing the original circle and its corresponding linear transformation in the w-plane.